

Supporting information

The Cyanotype Process and its Potential in Chemistry Education

Alexander Kmet' *, Anna Drozdíková, Soňa Nagyová, Peter Ikhardt

Faculty of Natural Sciences, Comenius University in Bratislava, Department of
Didactics in Science, Psychology and Pedagogy, Ilkovicova 6, Bratislava, Slovakia 84215

* kmet45@uniba.sk

Making cyanotypes – the blue photographs

A laboratory experiment for elementary-school students: The instructor notes for ISCED 2

List of all chemicals:

- Water
- iron(III) sulphate (CAS 10028-22-5)
- iron(II) sulphate (CAS 7720-78-7)
- ferric ammonium citrate (CAS 1185-57-5)
- potassium ferricyanide (CAS 13746-66-2)

Hazards:

Ferric ammonium citrate and iron(II) sulphate are hazardous in the case of skin and eye contact (irritant). Iron(III) sulphate is hazardous when swallowed and in case of skin and eye contact.

Ferric ammonium citrate is quite an inconvenient substance to work with, as it quickly absorbs humidity from the air, forming a sticky mass.

Every student should wear a lab coat, protective goggles, and gloves. No eating nor drinking is allowed during the lesson.

Introduction:

Some compounds undergo chemical changes when exposed to light, which makes them useful in the process of analogue photography. One of the compounds is ferric ammonium citrate. When mixed with potassium ferricyanide it becomes light-sensitive. After treating paper with this mixture and exposing it to sunlight, **cyanotype** – a blue photograph is produced. One of the first photography pioneers, Anna Atkins used this process to make cyanotypic photograms of plants: a procedure you'll be following during this lesson.

Demonstration of Prussian blue formation (≈ 10 minutes, performed by the teacher)

Equipment: 3 small beakers, marker

Chemicals: 5 % iron(III) sulphate solution, 5 % iron(II) sulphate solution, 5 % potassium ferricyanide solution

Demonstration: Mark the beakers "iron(III)", "iron(II)", and "ferricyanide" and add the solutions. Add few drops of ferricyanide into both other beakers and observe. The Prussian blue forms in Iron (II) sulphate solution only. The instructor explains, that some iron(III) salts can be reduced by sun, which is used in photography.

Making of cyanotypes (≈ 30 minutes, performed by the students)

Equipment: 2 small beakers, filtration paper, clip-in frame, brush, scissors, stirring rod, filtration paper

Chemicals: the instructor prepares solutions and mark the beakers according to the table S1 prior to the lesson:

Table S1. The preparation of the solutions

Beaker	A	B
Water	250 ml	250 ml
Substance	25 g of ferric ammonium citrate	35 g of potassium ferricyanide

Experimental procedure:

1. Mix the solutions A and B.
2. Spread the mixed solutions onto the paper using a brush.
3. Arrange plants on the paper and put it into the clip-in frame.
4. Take the frames outside and expose them to sunlight for 4 minutes. The paper should be dark blue after the exposure.
5. Wash the paper under running water until yellow areas become white.

Practical notes for the procedure:

- Step 2:* The paper should be wet, but not soaked and have yellow colour. There is no need to dry papers if we're gonna use them immediately.
- Step 4:* There's no need to adjust the exposure time in cloudy or sunny days, since there's no gradient in photograms (the contrast is represented only by exposed and unexposed areas). The 4-minute exposure has been efficient regardless the weather.
- Step 5:* Washing needs to be careful, since some papers tend to break under running water.